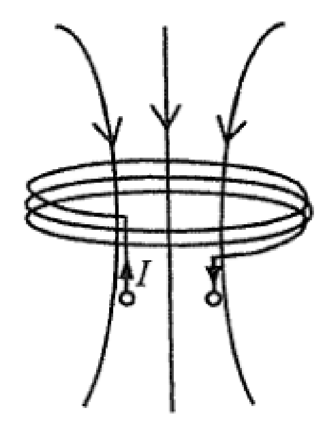
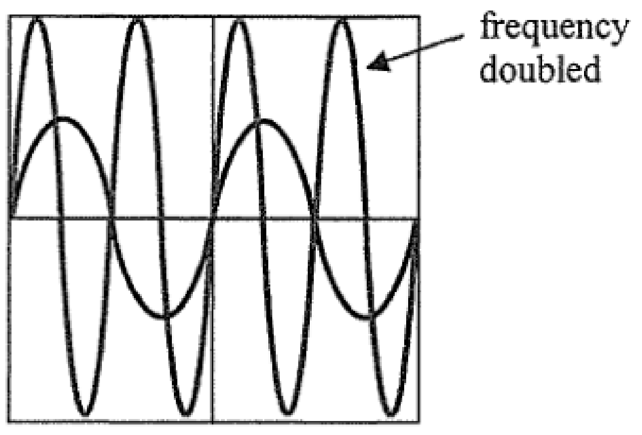
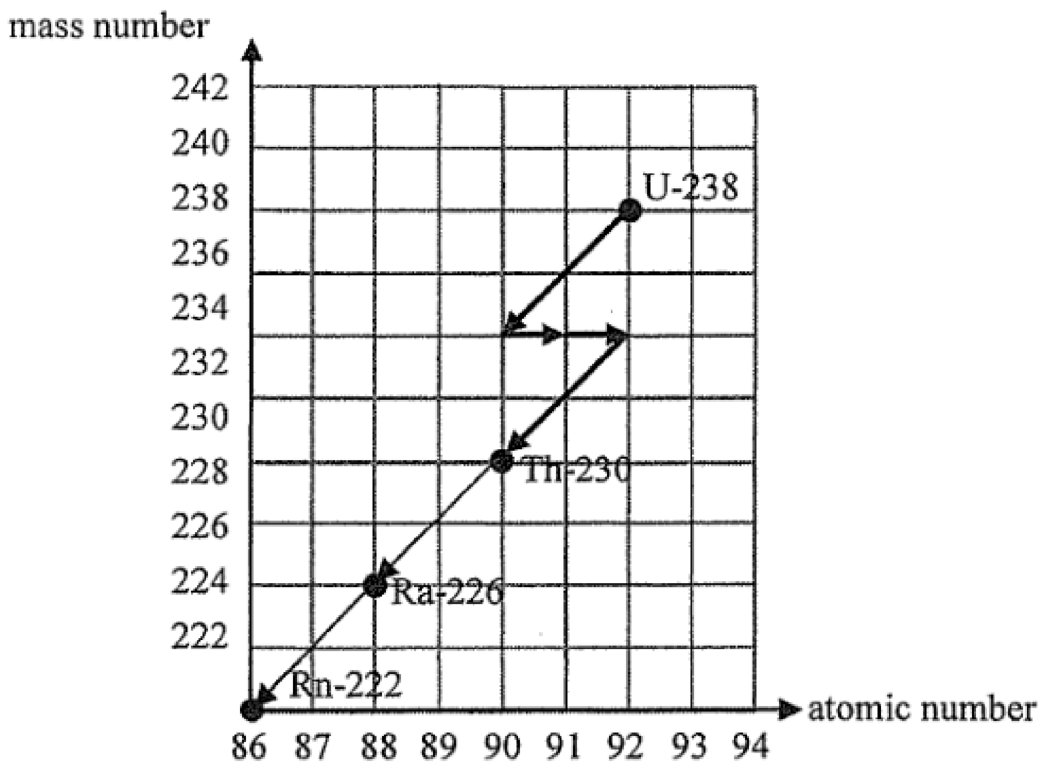
6.	(a)	(i)	Accelerates at g before the elastic cord stretches / at the beginning.	1A	
			Acceleration decreases as the cord stretches.	1A	
			Decelerates until momentarily at rest	1A	
			(after the tension in the cord is greater than mg).		$\underline{3}$
		(ii)	Gravitational potential energy changed to kinetic energy and	1A	
			(then) elastic potential energy in elastic cord.	1A	$\underline{2}$
	(b)		Elastic cord lengthens the stopping time,	1A	
			hence reduces the (net) force acting on the player.	1A	$\underline{2}$
	(c)		Contact area is larger,	1A	
			hence pressure is smaller during the fall and the structure is less likely to break / detach.	1A	$\underline{2}$

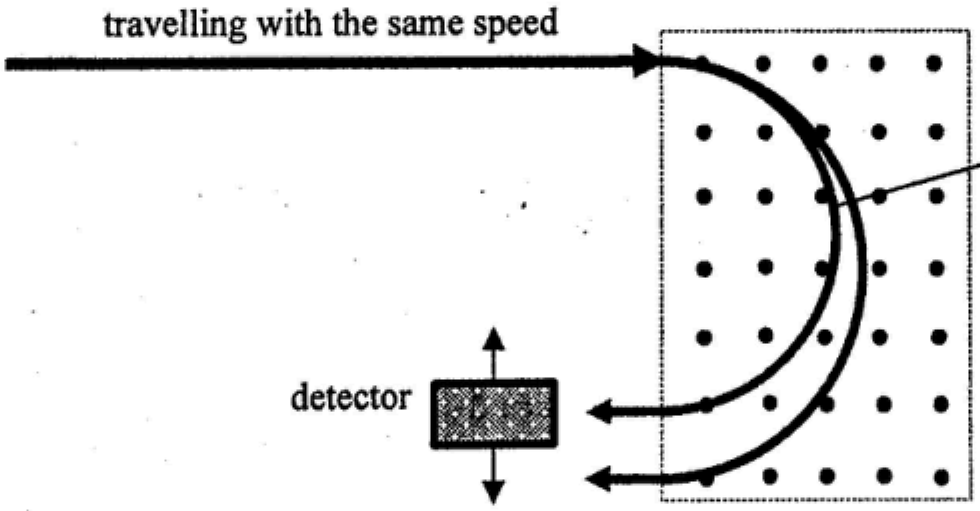
10. (a) Alpha particles emitted can be stopped by the (thin) metallic casing.	1A	
Or Shorter range / Lower penetrating power.	1A	
		1
(b) (i) $k = \dfrac{\ln 2}{t_{1/2}} = \dfrac{\ln 2}{87.74 \times 3.16 \times 10^7}$	1M	
$= 2.5 \times 10^{-10}\,\text{s}^{-1}\text{or}\,7.9 \times 10^{-3}\,\text{year}^{-1}$		
Activity $A = kN$		
$= \dfrac{\ln 2}{87.74 \times 3.16 \times 10^7}(3.2 \times 10^{25})$	1M	
$= 8.000 \times 10^{15}$ (Bq)	1A	
		3
(ii) Power = Energy per decay × Activity		
$= 5.5$ MeV $\times 8.000 \times 10^{15}$ Bq	1M	
$= 5.5 \times 10^6 \times 1.60 \times 10^{-19} \times 8.000 \times 10^{15}$		
$= 7040$ W or 7.040 (kW)	1A	
		2
(iii) Power $\propto$ Activity		
Activity $\propto N$		
$\therefore$ Percentage of power left $= (\frac{1}{2})^{t/t_{1/2}} \times 100\%$		
$= (\frac{1}{2})^{36/87.74} \times 100\%$	1M	
$= 75.25\% \approx 75\%$	1A	
Or Percentage / fraction of power left = 3/4		
Alternatively: $N = N_0 e^{-kt}$		
$\therefore$ Percentage of power left $= e^{-kt} \times 100\%$	1M	
$= e^{-(\ln 2 \div 87.74) \times 36} \times 100\%$		
$= e^{-0.2844} \times 100\%$		
$= 75.25\% \approx 75\%$	1A	
		2

10. (a) (i)	$226 - 206 = 20$ (multiple of 4 for α) $\therefore {}^{206}_{82}\text{Pb}$ is the end product	1M 1A	2	
(ii)	% undecayed Ra-226 left $\dfrac{N}{N_0} = e^{-\frac{\ell n2}{1600}\times 50}$ $= 97.857\ \%$ $\approx 97.9\ \%$	1M 1A		Accept 97.8 % to 98.0 %
	Or % undecayed Ra-226 left $= \left(\dfrac{1}{2}\right)^{\frac{50}{1600}}$ $\approx 97.9\ \%$	1M 1A	2	
(b) (i)	$\because$ random process	1A	1	
(ii)	Some of the daughter products of Ra-226 are also radioactive and may emit β particles.	1A	1	
(iii)	Reason: weaker ionizing power of β and γ radiations - raise the source to a distance greater than the range of α (a few cm) will cease to have sparks. - insert a paper between the source and the gauze, sparks will cease to produce. Any ONE	1A 1A	2	

9.	(a)	(i)	β decay / beta decay OR $^{40}_{19}\text{K} \rightarrow {}^{40}_{20}\text{Ca} + {}^{0}_{-1}\beta$	1A 1	
		(ii)	Justified: the penetrating power of β radiation enables it to penetrate the body's organ / skin Or Not justified: the activity is low and is comparable to background radiation / β radiation is largely shielded by the human body	1A 1	
	(b)	(i)	$\frac{0.45 \times 0.012\%}{40.0} = 1.35 \times 10^{-6}$ (mole)	1A 1	
		(ii)	$k = \frac{\ln 2}{1.25 \times 10^{9}\ (3.16 \times 10^{7})} = 1.754803 \times 10^{-17}\ (\text{s}^{-1})$ Activity = kN $= 1.754803 \times 10^{-17} \times (1.35 \times 10^{-6} \times 6.02 \times 10^{23})$ $= 14.261284$ (Bq) ≈ 14.3 (Bq)	1M/1A 1A 2	i. e. $5.545177 \times 10^{-10}\ (\text{year}^{-1})$ Accept 14.2 to 14.3 (Bq)

12. (a) 	2A	1A for correct direction (also accept field lines pointing upwards for observers with different perception of current direction) 1A for correct field pattern
	2	
(b) (i) The a.c. flowing in transmitter coil T generates a changing magnetic field,	1A	
thus by electromagnetic induction an induced e.m.f. is produced in receiver coil R to oppose the changing magnetic flux experienced by it.	1A	
	2	
(ii) CRO display 	2A	1A for correct frequency 1A for correct amplitude
	2	
(c) If the back cover is metallic, eddy currents will be produced (by induction),	1A	
loss of energy / heating effect results / magnetic field / flux is blocked by the metal case and thus making wireless charging impossible. Or Non-metallic materials such as glass are insulators, and no eddy currents are produced. As a result, energy loss / heating effect will be minimized / the magnetic field can pass through the back cover easily.	1A	
	2	

Solution	Marks	Remarks
13. (a) (i) $k = \frac{\ln 2}{t_{½}}$		
$= \frac{\ln 2}{3.82 \times 24 \times 3600}$	1M	
$= 2.1001405 \times 10^{-6}\ \text{s}^{-1} \approx 2.10 \times 10^{-6}\ (\text{s}^{-1})$	1A	
	2	
(ii) $A = kN$		
$N = \frac{48}{2.10 \times 10^{-6}}$	1M	e.c.f. from (a)(i)
$= 2.285561 \times 10^{7} \approx 2.29 \times 10^{7}$	1A	Accept: $(2.28 \sim 2.3) \times 10^{7}$
	2	
(iii) As radon is a gas that can be inhaled into the lungs of human,	1A	
the relatively strong ionizing power of α particles emitted by decaying radon / radioactive gas might affect the organs / cells nearby.	1A	
	2	
(b) 	2A	
	2	

Solution	Marks	Remarks
12. (a) (i) $^{14}_{6}C \rightarrow ^{14}_{7}N + ^{0}_{-1}e$ (or β)	1A	
	1	
(ii) $k = \dfrac{\ln 2}{t_{\frac{1}{2}}}$		
$= \dfrac{\ln 2}{5370}$	1M	
$= 1.2097 \times 10^{-4}$ (yr^{-1}) $\approx 1.21 \times 10^{-4}$ (yr^{-1})	1A	
	2	
(iii) $C = C_0 e^{-kt}$		
$6.0 = 13.6\, e^{-(1.21\times10^{-4})t}$	1M	e.c.f. from (a)(ii)
$t = 6764.7 \approx 6800$ (years)	1A	Accept: 6760 ~ 6820 (years)
		Or $(13.6)\left(\frac{1}{2}\right)^n = 6.0$ $n\log\left(\frac{1}{2}\right) = \log\left(\frac{6.0}{13.6}\right)$ $n = 1.181$ $t = n \times 5730 = 6764.7$ (years) ≈ 6800 (years)
	2	
(b) (i) An isotope is a variation of an element that possesses the same atomic number but a different mass number.	1A	Or same number of protons but different number of neutrons
	1	
(ii) 	1A	
	1	
(iii) As the magnetic force induced on a carbon ion is always perpendicular to its velocity / direction of motion,	1A	
no work is done on the ions.	1A	
	2	